

Roll No.

TPE-301

B. TECH. (PE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

THERMODYNAMICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) State whether the following properties are intensive or extensive :

(CO1, CO2)

- (i) Volume
- (ii) Density
- (iii) Specific Volume
- (iv) Heat Capacity
- (v) Specific Heat
- (vi) Potential Energy
- (vii) Pressure
- (viii) Temperature
- (ix) Specific Heat Capacity
- (x) Specific Gravity

P. T. O.

OR

(b) Define second law of thermodynamics. (CO1, CO2)

2. (a) A volume of 0.36 m^3 of gas at 1.03 bar and 288 K is compressed reversible and adiabatically to 10 bar. It is then cooled at constant pressure to its original temperature after which it expands isothermally. Calculate the following : (CO2)

- (i) The heat transfer at constant pressure
- (ii) The heat transfer during expansion
- (iii) The total work done during the cycle.

OR

- (b) Prove $PV^\gamma = \text{constant}$ for the adiabatic process and hence derive this expression for work done by adiabatic process i.e.

$$W = \frac{P_1 V_1}{\gamma - 1} \left[1 - \frac{P_2^\gamma}{P_1} \right]. \quad (\text{CO2})$$

3. (a) Why van der Waals equation to be studied ? Derive the constant values of 'a' and 'b' in terms of critical temperature and pressure. (CO1)

OR

- (b) Air at 1.02 bar, 22°C , initially occupying a cylinder volume of 0.015 m^3 , is compressed reversibly and adiabatically by a piston to a pressure of 6.8 bar. Calculate : (CO1)

- (i) The final temperature
- (ii) The final volume
- (iii) The work done

4. (a) Establish the relationship between PV, TV and PT for an adiabatic process. (CO1, CO2)

(3)

OR

- (b) Explain the Carnot principle. Draw the same on the P-V diagram. Explain the different processes involved in the principle. (CO1, CO2)

5. (a) Discuss the equivalence of the Kelvin and Clausius statement. Hence solve :

A heat engine receives heat at the rate of 1500 kJ/min. and gives an output of 8.2 kW. (CO1, CO2)

Determine :

- (i) The thermal efficiency.
- (ii) The rate of heat rejection

OR

- (b) Discuss and explain the entropy and the nature of the process in second law of thermodynamics. (CO1, CO2)